

OLS - Ordinary least Squares

x	y
1	2
2	4
3	5
4	8

$$\hat{y} = \beta_0 + \beta_1 x$$

}

Residual error:

$$e_i = y_i - \hat{y}_i$$

if

$$\hat{y} = 0.5 + 1.8x$$

$$\begin{aligned} x &= 3 \\ \hat{y} &= 5.9 \\ y &= 5 \end{aligned}$$

$$\Rightarrow e = 5 - 5.9 = -0.9$$

- OLS asks which values of β_0, β_1 give us the "best" linear fit to the observed data.

$$\min \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

derivative with respect to β_0

$$f(\beta_0, \beta_1) = \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2$$

$$\frac{\partial f}{\partial \beta_0} = -2 \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i) = 0$$

$$\sum (y_i - \beta_0 - \beta_1 x_i) = 0$$

$$\sum y_i - n\beta_0 - \beta_1 \sum x_i = 0 \Rightarrow n\beta_0 + \beta_1 \sum x_i = \sum y_i$$